Week 13 Lecture 1 STAT2004/2904/7004

- Ass 4 due Fri 30 Oct @ 4pm
- Ross' review lecture on Friday
- Alan's review session during SWOTVAC

Recap:

(1) A linear model covers any model of the form

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon},$$

A linear model is linear in its regression coefficients β.

(2) For hypothesis testing about $\underset{\sim}{\beta}$ in linear models, i.e.;

$H_0: R\underset{\sim}{\beta} = 0$ versus $H_1: R\underset{\sim}{\beta} \neq 0$ for some matrix R with rank$(R) = r$,

the GLRT reduces to the following ANOVA

	Model	df	RSS	F	$P_r(F)$
"Effect"	<difference>	r	$RSS_0 - RSS_1$	$\frac{RSS_0 - RSS_1 / r}{RSS_1 / n-p}$	$P_r(F_{r,n-p} \geq F)$
"Residual"	Alternative	$n-p$	RSS_1		
"Total"	Null	$n-(p-r)$	RSS_0		

Example 8.8 (Jump distances, continued)

A starting model here was

$$jump_i = \beta_0 + \beta_1 height_i + \beta_2 height_i^2 + \beta_3 height_i^3 + \varepsilon_i$$

Q: Do we really need a cubic model in height to explain jump distance, or is a linear model enough?

[R DEMO]